

# Harjot Singh Raith

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## EDUCATION

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|------------------------------------------------------------------------------|-----------------------------------|
| Narsee Monjee Institute of Management and Studies(NMIMS), Navi Mumbai        | July 2022 – April 2026            |
| B.Tech in Artificial Intelligence and Data Science                           | CGPA: 3.2/4.0                     |
| Shubham Raje Junior College, Thane                                           | June 2020 – May 2022              |
| Maharashtra State Board of Secondary and Higher Secondary Education, Science | Grade (12 <sup>th</sup> ): 85.83% |
| New Horizon Scholars School, Thane                                           | May 2015 – April 2020             |
| Central Board of Secondary Education (CBSE)                                  | Grade (10 <sup>th</sup> ): 90.40% |

## RESEARCH & WORK EXPERIENCE

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| <b>AI Backend Engineer (Intern)</b><br>Paloma POS, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Dec 2025 – Present<br>Remote              |
| <ul style="list-style-type: none"><li>Designed and implemented end-to-end production-grade <b>RAG pipelines</b> using Dify workflows, integrated with a centralized <b>AI Gateway (Golang)</b>; implemented low-latency, <b>scalable semantic search</b> using <b>OpenAI embeddings</b> and <b>Qdrant</b>, managing large-scale vector data and retrieval.</li><li>Designed robust, <b>multi-tenant knowledge base ingestion</b> and AI data infrastructure, including automated web scraping, document chunking, embedding lifecycle management, and secure data isolation.</li></ul>    |                                           |
| <b>e-Yantra Research Intern at IIT Bombay</b><br>Embedded Real-Time Systems Laboratory (eRTS Lab), IIT Bombay                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | May 2025 – July 2025<br>IIT Bombay, India |
| <ul style="list-style-type: none"><li>Developed a <b>full-stack AI-driven Thematic analysis platform</b> for multi-format document processing with automated inductive and deductive coding. Built a <b>FastAPI–PostgreSQL backend</b> integrated with <b>LLM-based agents</b> to generate codes, themes, and structured research reports.</li><li>Designed a <b>React/MUI frontend</b> with interactive visualizations, including <b>Sankey diagrams</b>, delivering an end-to-end <b>Qualitative Research tool</b> for accelerated insight extraction.</li></ul>                        |                                           |
| <b>e-Yantra Research Intern at IIT Bombay</b> <a href="#">Certificate</a><br>Embedded Real-Time Systems Laboratory (eRTS Lab), IIT Bombay                                                                                                                                                                                                                                                                                                                                                                                                                                                 | May 2024 – July 2024<br>IIT Bombay, India |
| <ul style="list-style-type: none"><li>Analyzed large-scale user interaction logs using <b>Python</b>, leveraging <b>Google OCR</b> for video log generation and <b>CNNs</b> to detect and filter low-quality videos.</li><li>Collected, cleaned, and integrated log data via <b>REST APIs</b> from a <b>Laravel-based platform</b> to measure scaffold availability and user dwell time.</li><li>Applied <b>data visualization and behavioral analytics</b> to optimize scaffolding and user engagement using clustering, heatmaps, sequence analysis, and predictive modeling.</li></ul> |                                           |

## RESEARCH PUBLICATION

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Enhancing and analyzing log generation for Collaborative Problem-Solving activities: Video analysis and OCR techniques</b> — Published in Annual ACM India Compute Conference (Compute 2024), CCIS vol. 2400, Springer Nature. | <a href="#">Research Paper Link</a> |
| <ul style="list-style-type: none"><li>Suprabha Jadhav, Parth Jain, <b>Harjot Singh Raith</b>, Sridhar Iyer, Kavi Arya</li></ul>                                                                                                   |                                     |
| <b>YTubeRAG: Leveraging YouTube Content for Enhanced LLM Training with Optimized Vector Space Retrieval</b> — Presented in NTAI 2025, CRC Press (Taylor & Francis Group), Scopus-indexed international conference, 2025.          | <a href="#">Paper Certificate</a>   |
| <ul style="list-style-type: none"><li><b>Harjot Singh Raith</b>, Kartikeya Mudliyar, Ishika Mohan, Aditya Kasar, Sakshi Indolia</li></ul>                                                                                         |                                     |
| <b>AI-Driven Timetable Generation and Optimization using Constraint-Based Scheduling</b> — IEEE Conference Proceedings, ICESIC (International Conference on Emerging Systems and Intelligent Computing), 2026.                    | Accepted                            |
| <ul style="list-style-type: none"><li><b>Harjot Singh Raith</b>, Kartikeya Mudliyar, Toral Shah</li></ul>                                                                                                                         |                                     |

## PROJECTS

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| <b>LiveWire: Real-time code compiler</b> <a href="#">Github</a> <a href="#">Website</a> <a href="#">Youtube</a>                                                                                                                                                                                                                                                                                    | (AWS Global Vibe) |
| <ul style="list-style-type: none"><li>Built a real-time code-compiler platform for developers using Monaco, WebSockets, and Gemini AI.</li><li>Tools Used: TypeScript, Vite, Tailwind CSS, Monaco Editor, Socket.IO, Express, Node.js, CodeMirror, Vercel, Railway.</li></ul>                                                                                                                      |                   |
| <b>ArgoMindAI: Argo Float Dashboard &amp; Mapbot</b> <a href="#">Github</a> <a href="#">Website</a> <a href="#">Youtube</a>                                                                                                                                                                                                                                                                        | (SIH'25)          |
| <ul style="list-style-type: none"><li>Developed an automated pipeline to ingest FTP raw files, store them in S3, and generate a sequential schema for real-time dashboards and a map-based RAG system using Argo data with ML-driven ocean disaster predictions.</li><li>Tools Used: React, Python (UV), FastAPI, Xarray, Tailwind CSS, Leaflet.js, AWS S3, AlloyDB, Docker, Gemini API.</li></ul> |                   |

**AlmanacAI: Timetable generation & optimization**  [Github](#)  [Website](#)  [Youtube](#) (Capstone Project)

- Created a GA/CSP/backtracking-based optimized timetable generator with RAG for quick class and faculty lookup.
- Tools Used: React, Vite, Tailwind CSS, MUI, Axios, Node.js, Express.js, MongoDB, Gemini AI, Vercel, Render.

**Biros: Bio-Inspired Robotic Snake**  [Github](#)  [Youtube](#) (Gujarat Robofest'25)

- Created an autonomous robotic snake using CNNs, kinematics, and LiDAR for detection in confined and disaster areas.
- Tools Used: ESP32, Servo Motor MG995, VL53L0X, LiDAR Sensors, OpenCV, Inverse and Forward Kinematics.

## ACHIEVEMENTS & AWARDS

**2nd Place — VOIS International Hackathon 2024 by Vodafone** [Certificate](#)

Secured 2<sup>nd</sup> place out of 8000+ teams from India, Romania & Egypt • Awarded €4,000

**Judges' Choice Award — eYIC 2023-24 Finals, IIT Bombay** [Certificate](#)

Competed out of 299 teams in the finale • Awarded INR 50,000

**Top 10 Finalist — ROBOFEST GUJARAT 4.0 COMPETITION, Science City Ahmedabad** [Certificate](#)

Top 10 teams out of 5000+ teams • Awarded INR 2,50,000 for proof-of-concept and prototyping

**3rd Place — Smart Innovation Model Challenge (BASIC 4.0), VES Business School** [Certificate](#)

Secured 3<sup>rd</sup> place out of 50+ teams • Awarded INR 15,000

**3rd Place — Startup Pitch Competition (BASIC 4.0), VES Business School** [Certificate](#)

Secured 3<sup>rd</sup> place out of 80+ teams • Awarded INR 15,000

**Consolation Prize — 9th Multidisciplinary International Conference, Mumbai** [Certificate](#)

Paper on LLMs for Indian vernacular speech recognition

**Multiple Hackathon Wins** [All Certificates](#)

30+ hackathons • 8 wins + numerous awards

## TECHNICAL SKILLS

**Languages:** Python, CUDA, Shell, LaTeX, C, C++, Java, SQL, JavaScript, TypeScript.

**Dev Tools:** Linux, AWS, GCP, Docker, Kubernetes, Hugging Face, Wandb, Overleaf, GitHub.

**Frameworks:** TensorFlow, PyTorch, FastAPI, Flask, OpenCV, Langchain, Streamlit, Keras, Selenium.

**Relevant Coursework:** Deep Learning, Machine Learning, Discrete Mathematics, Probability and Statistics, Computer Networks, Cryptography and Network Security, Data Structures & Algorithms and Object Oriented Programming (OOP).

## COURSES & CERTIFICATIONS

**Cloud Computing — NPTEL, IIT Kharagpur** [Certificate](#)

**Generative AI Language Modeling with Transformers — IBM** [Certificate](#)

**Introduction to Generative AI — Google Cloud** [Certificate](#)

**Applied ChatGPT for Cybersecurity — Infosec** [Certificate](#)

**AI Workflow: Enterprise Model Deployment — IBM** [Certificate](#)

**Foundation of AI and Machine Learning — Microsoft** [Certificate](#)

**Python Fundamentals — DataCamp** [Certificate](#)

**SQL Fundamentals — DataCamp** [Certificate](#)

## TEACHING EXPERIENCE

**Technical Workshop Instructor — NMIMS Robotics Club, Navi Mumbai** **July 2024 - July 2025**

- Conducted workshops on Git, embedded systems, full-stack architecture, and ML-based RAG pipelines for all engineering students .
- Organized "Open Source to Deployment: A Full-Stack Journey" workshop series, providing hands-on mentorship in robotics, applied AI, and deployment workflows. Promoted open-source contribution and collaborative development through technical training sessions

**Mathematics & Science Tutor — Ma Niketan NGO, Thane** **Volunteer**

- Taught mathematics and science to underprivileged students, focusing on foundational concepts and problem-solving skills. Developed engaging lesson plans to improve student comprehension and academic performance in STEM subjects

## EXTRACURRICULAR ACTIVITIES

**Head of the Robotics Club — NMIMS Navi Mumbai** **July 2024 - July 2025**

- Led a team of 50+ members in organizing technical workshops and robotics competitions; coordinated inter-college events to promote STEM education and hands-on learning experiences

**Core Member of the Manthan Club — NMIMS Navi Mumbai** **Sept 2022 - Jan 2023**

- Collaborated with team members to organize cultural and technical events; contributed to event planning, logistics, and execution to enhance student engagement and participation